

HAEJU, Korea, Democratic People's Republic ofWMO: **470690**Lat: **38.033N**Lon: **125.700E**Elev: **266**StdP: **14.56**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	15.2	-4.0	4.5	18.1	-0.4	5.4	18.8	24.9	23.2	21.4	25.2	7.4	320	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.3	88.5	78.4	85.9	76.5	83.7	75.4	80.3	86.1	78.9	83.8	77.6	81.7	5.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	150.5	83.9	77.4	144.1	82.0	76.3	138.6	80.7	44.2	86.4	42.7	83.8	41.4	81.8	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.2	15.3	7.6	91.7	4.6	2.1	4.4	93.2	1.7	94.4	-0.9	95.6	-4.2	97.1		
(5)			5.6	82.4	4.1	2.0	2.7	83.8	0.2	84.9	-2.1	86.0	-5.1	87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	27.1	31.7	40.9	52.6	62.4	70.8	75.9	78.1	70.5	59.2	44.4	31.8	(6)
(7)		DBStd	18.46	6.88	6.60	5.85	5.95	4.95	3.95	3.78	4.06	4.29	5.92	7.93	7.21	(7)
(8)		HDD50	2321	708	514	288	38	0	0	0	0	0	8	201	564	(8)
(9)		HDD65	5192	1173	934	748	374	111	5	0	8	192	618	1029		(9)
(10)		CDD50	3743	0	0	4	116	385	624	804	870	614	293	33	0	(10)
(11)		CDD65	1140	0	0	0	1	31	179	339	406	172	12	0	0	(11)
(12)		CDH74	7116	0	0	0	5	164	793	2057	3367	702	28	0	0	(12)
(13)		CDH80	1638	0	0	0	0	16	141	461	943	77	0	0	0	(13)
(14)	Wind	WSAvg	5.1	5.2	5.6	6.1	5.9	5.4	4.5	4.7	5.0	4.3	4.6	5.0	5.1	(14)
(15)	Precipitation	PrecAvg	41.9	0.4	0.7	1.0	2.0	4.0	4.6	12.6	9.2	4.3	1.7	1.5	0.6	(15)
(16)		PrecMax	80.8	1.2	2.3	3.5	4.8	9.3	18.2	29.1	19.7	18.2	5.6	3.5	1.7	(16)
(17)		PrecMin	19.8	0.0	0.0	0.0	0.0	0.3	0.6	3.8	1.5	0.5	0.1	0.2	0.1	(17)
(18)		PrecStd	12.4	0.3	0.6	0.8	1.1	2.4	3.5	6.1	5.0	4.3	1.5	0.9	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	52.4	61.3	74.1	82.2	87.5	91.5	92.3	85.1	77.7	65.3	51.2	(19)
MCWB			42.6	45.8	50.5	60.5	65.9	71.8	79.4	81.3	73.7	66.0	57.1	46.1	(20)	
2%		DB	42.0	47.5	56.1	69.3	78.2	83.8	87.4	89.7	82.1	73.5	61.2	47.7	(21)	
		MCWB	37.6	40.6	46.9	56.4	63.8	70.4	78.1	80.3	72.4	64.0	55.9	43.4	(22)	
5%		DB	39.5	43.8	52.9	65.5	75.1	80.9	84.5	87.2	80.1	70.8	58.4	45.0	(23)	
		MCWB	35.4	38.2	44.9	54.2	62.7	69.6	76.7	78.1	70.7	62.3	53.8	41.1	(24)	
10%		DB	36.9	41.0	50.0	62.4	71.8	78.5	82.3	85.0	78.1	68.3	55.8	42.4	(25)	
		MCWB	33.2	36.4	43.5	52.9	61.4	68.9	75.7	77.0	69.2	60.7	51.0	38.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	47.8	53.1	63.7	70.1	76.0	81.7	83.1	77.0	68.8	60.9	47.7	(27)
MCD B			45.9	50.3	57.8	71.5	77.6	82.9	87.6	90.2	81.6	74.4	62.7	51.0	(28)	
2%		WB	38.8	41.9	49.2	58.8	66.7	73.2	79.5	81.0	74.8	66.1	57.3	44.4	(29)	
		MCD B	41.3	45.6	53.8	64.8	74.8	79.9	85.2	87.2	79.4	71.2	59.7	47.0	(30)	
5%		WB	36.0	39.2	46.3	56.5	64.7	71.8	78.0	79.8	73.0	64.1	54.5	41.5	(31)	
		MCD B	38.9	42.6	51.0	63.1	71.9	77.7	82.7	85.1	77.8	68.8	57.7	44.6	(32)	
10%		WB	33.6	37.0	44.2	54.4	62.9	70.5	76.7	78.7	71.3	62.1	51.4	38.8	(33)	
		MCD B	36.5	40.5	49.1	60.9	69.2	75.9	80.5	83.2	76.3	67.0	55.4	42.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.0	12.8	13.6	14.2	14.1	12.0	8.5	10.3	12.9	14.4	12.3	11.5	(35)
MCD BR			13.0	14.8	17.0	18.1	19.5	17.0	12.9	13.0	15.2	16.3	14.2	13.1	(36)	
MCWBR			10.4	11.3	11.4	10.5	10.6	9.3	7.7	7.0	8.6	10.1	10.8	11.1	(37)	
5% WB		MCD BR	12.0	13.1	14.8	15.1	15.9	13.0	10.5	11.0	12.8	14.1	12.4	12.4	(38)	
		MCWBR	10.8	11.3	11.6	10.3	10.4	8.3	7.0	6.5	8.5	10.2	11.0	11.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.425	0.485	0.553	0.607	0.621	0.701	0.671	0.578	0.510	0.514	0.492	0.432	(40)	
taud		1.935	1.830	1.720	1.656	1.676	1.587	1.697	1.905	1.997	1.870	1.867	1.940	(41)		
Ebn at Noon		222	225	224	222	222	205	209	227	234	214	198	208	(42)		
Edh at Noon		44	56	69	77	77	84	75	60	51	53	47	41	(43)		
(44)	All-Sky Solar Radiation	RadAvg	760	1018	1378	1597	1777	1639	1321	1421	1372	1138	757	656	(44)	
(45)		RadStd	55	74	116	107	166	179	189	184	127	75	77	41	(45)	